Announcements

Topic 5: Review

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Step Counting

```
int size = array.length;
for (int i = 0; i < size - 1; i++):
    for (int j = 0; j < size - i - 1; j++):
        if (array[j] > array[j + 1]):
            int temp = array[j]
            array[j] = array[j + 1]
            array[j + 1] = temp
```

- Outside for loops: 3
- Outer for loop: 4
- Inner for loop: 13
- Now… How to combine the for loops?

```
int size = array.length, o++ # variable assignment
o++ # outer for loop initialization
for (int i = 0; i < size - 1; i++):
        o++ # outer for loop true conditional evaluation
        o++ # inner for loop initialization
        for (int j = 0; j < size - i - 1; j++):
                o++ # inner for loop true conditional evaluation
                if (array[j] > array[j + 1]): (o++ * 6)
                        int temp = array[j] (o++)
                        array[j] = array[j + 1] (o++ * 2)
                        array[j + 1] = temp (o++ * 2)
                o++ # inner for loop incremenation
        o++; # inner for loop false conditional evaluation
        o++; # outer loop incrementation
o++; # outer for loop false conditional evaluation
```

Step Counting

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int size = array.length;
for (int i = 0; i < size - 1; i++):
    for (int j = 0; j < size - i - 1; j++):
        if (array[j] > array[j + 1]):
            int temp = array[j]
            array[j] = array[j + 1]
            array[j + 1] = temp
```

- Outside for loops: 3, Outer for loop: 4, Inner for loop: 13
- The inner for loop goes from 0 to size (n) – i – 2 (n – i – 1 total iterations) where i starts at 0 and goes to n – 2 (n – 1 total iterations).
- We have to separate out the outer loop's iteration from the inner loop, using a summation for the inner loop. How many times will the inner loop run?
- $\sum_{i=0}^{n-2} n - i - 1$ is the summation for the inner loop. This is the total number of times it will run, so we just add it to the outer loop operations instead of multiplying it

Step Counting

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for (int i = 0; i < size - 1; i++):
    for (int j = 0; j < size - i - 1; j++):
        if (array[j] > array[j + 1]):
            int temp = array[j]
            array[j] = array[j + 1]
            array[j + 1] = temp
```

- Outside for loops: 3, Outer for loop: 4, Inner for loop: 13, total number of inner loop iterations: $\sum_{i=0}^{n-2} n - i - 1$, total number of outer loop iterations: 4(n – 1).
- Combining it all together, we get: 3 + 4(n – 1) + $13 * \sum_{i=0}^{n-2} n - i - 1$.